Original Article

Spin-Encoded Reactivity in Monovalent Boron: Substituent-Imprinted Singlet-Triplet Signatures of Borylenes

Mehdi Nabati*

Independent Researcher (Computational Chemistry and Drug Design), Tehran, Iran

Use your device to scan and read the article online

Citation M. Nabati, **Spin-Encoded Reactivity in Monovalent Boron: Substituent-Imprinted Singlet-Triplet Signatures of Borylenes.** *Eurasian J. Sci. Technol.*, **2026**, 6(4), 309-330.

https://doi.org/10.48309/ejst.2026.578182.1308

CC BY

Article info:
Received: 2026-02-26
Accepted: 2026-09-05
Available Online: 2026-09-11
ID: EJST-2602-1308

Keywords:
Singlet-triplet energy gap; Boron lone pair hybridization; σ_{B-X} bond energy; ^{11}B-NMR spectroscopy; Polarizability

ABSTRACT

A systematic quantum chemical investigation was carried out on sixteen monovalent B–X derivatives to elucidate substituent-controlled stabilization, electronic structure, and magnetic response in both singlet and triplet states. Calculated singlet–triplet energy gaps (ΔE_{S-T}) reveal that several substituents strongly favor singlet stabilization, with larger positive ΔE values directly correlating with enhanced thermodynamic stability. Analysis of σ_{B-X} bond energies demonstrates that stronger bonding interactions generally accompany increased singlet preference, establishing a clear energetic-structural relationship. Natural bond orbital (NBO) analysis shows that the boron lone pair (LP) in stabilized singlets possesses significant s-character, indicating pronounced electron localization rather than pure p-type behavior. Spin density calculations confirm efficient spin delocalization in triplet states, while singlets exhibit reduced magnetic perturbation at the boron center. Magnetic shielding calculations further reveal systematic variations in ^{11}B-NMR chemical shifts consistent with changes in electronic density and bonding strength. Additionally, computed dipole moments and isotropic/anisotropic polarizabilities demonstrate that substituents modulate not only stability, but also electronic deformability and charge distribution. Overall, this study establishes quantitative correlations between ΔE_{S-T}, σ_{B-X} bond strength, hybridization patterns, NMR response, dipole moments, and polarizability, providing a unified electronic framework for understanding substituent-driven stabilization in B-centered systems.

Introduction

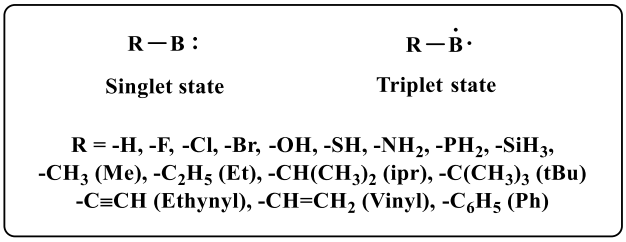
Boron-containing compounds occupy a distinctive position at the interface of fundamental chemistry and applied sciences. Due to their electron-deficient nature and versatile bonding patterns, boron derivatives have found widespread applications ranging from organic synthesis and materials science to medicine and pharmaceutical chemistry [1-3]. In particular, boron-based motifs play a pivotal role in boron neutron capture therapy (BNCT), enzyme inhibition strategies, and drug design, where their unique Lewis acidity and tunable electronic properties enable selective biological

*Corresponding Author: Mehdi Nabati; Email: mnabati@ymail.com

interactions [4-6]. Beyond medicinal chemistry, boron compounds are integral to advanced functional materials, catalysis, and optoelectronic devices, underscoring the broad technological relevance of understanding their electronic structure and reactivity at a fundamental level [7-9].

Among the diverse family of boron species, borylenes-neutral monovalent boron compounds of the general form R–B, represent some of the most intriguing and unconventional electronic systems. These species are characterized by a low coordination number and an unusual valence electron configuration, which allows them to exist in either singlet or triplet electronic states depending on substituent effects and electronic stabilization [10-12]. Conceptually analogous to carbenes, borylenes exhibit a delicate balance between lone-pair donation and vacant p-orbital acceptor character in the singlet state, while the triplet state features two unpaired electrons residing in nearly degenerate orbitals. Since their theoretical prediction and subsequent experimental and computational investigations, borylenes have attracted sustained interest as reactive intermediates and as platforms for probing spin-state-dependent bonding and reactivity in main-group chemistry [13-16].

This study presents a comprehensive density functional theory (DFT) investigation of sixteen distinct borylenes (Scheme 1) bearing a diverse set of substituents, encompassing hydride, halogen, heteroatom-containing, alkyl, alkenyl, alkynyl, and aryl groups. Both singlet and triplet electronic states are systematically examined to elucidate substituent-imprinted effects on molecular structure, thermodynamic stability, and intrinsic reactivity. Detailed analyses of molecular electrostatic potential (MEP) surfaces, NBO descriptors, and boron NMR parameters are employed to uncover clear spin-state signatures and electronic trends across the series. Through this multifaceted approach, the present study aims to establish a coherent structure-property-spin relationship for monovalent boron species and to provide deeper insight into the spin-encoded reactivity patterns of borylenes.



Scheme 1: All investigated borylenes in the present work

Computational Methods

All quantum chemical calculations reported in this work were carried out using the Gaussian 2016 suite of programs [17]. Geometry optimizations and electronic structure calculations were performed within the framework of DFT employing the B3LYP hybrid exchange-correlation functional in conjunction with the 6-311++G(d,p) basis set for all atoms. This level of theory has been shown to provide a reliable balance between computational efficiency and accuracy for main-group elements, particularly for boron-containing systems exhibiting delicate electronic features and multiple spin states.

A series of sixteen monovalent borylenes of the general formula R-B were systematically investigated, where R = H, F, Cl, Br, OH, SH, NH_2, PH_2, SiH_3, CH_3 (Me), C_2H_5 (Et), $CH(CH_3)_2$ (*i*Pr), $C(CH_3)_3$ (*t*Bu), C≡CH (ethynyl), $CH=CH_2$ (vinyl), and C_6H_5 (phenyl).

For each derivative, both singlet and triplet electronic states were explicitly considered in order to assess substituent-dependent spin effects on structure, stability, and reactivity.

Optimized geometries were obtained without symmetry constraints, and the nature of the stationary points was verified by the absence of

imaginary frequencies. Electronic structure characteristics were analyzed through detailed examination of frontier molecular orbitals (FMOs), including HOMO-LUMO energies and gaps, providing insights into spin-state-dependent reactivity trends. Molecular polarity was evaluated via computed dipole moments, while static polarizabilities were calculated to assess the electronic responsiveness of the borylenes toward external electric fields.

To further quantify chemical reactivity, a set of global reactivity descriptors-such as chemical hardness, softness, electronegativity, and electrophilicity index-were derived from frontier orbital energies within the conceptual DFT framework (CDFT). MEP surfaces were generated to visualize charge distribution patterns and identify electrophilic and nucleophilic regions as a function of substituent and spin state.

Bonding characteristics and charge-transfer effects were explored using NBO analysis, enabling a detailed description of σ/π interactions, lone-pair donation, and hyperconjugative effects at the boron center. Additionally, ^{11}B-NMR shielding constants were computed to establish correlations between electronic structure, spin multiplicity, and experimentally relevant spectroscopic signatures. The combined use of electronic, energetic, and spectroscopic descriptors provides a coherent and comprehensive image of spin-encoded reactivity in monovalent borylenes.

Results and Discussion

Structural Characteristics of Singlet and Triplet of Monovalent Borylenes

Table 1 summarizes the optimized B–X bond lengths for a series of sixteen monovalent borylenes (R-B) in both their singlet and triplet electronic states, as obtained at the B3LYP/6-311++G(d,p) level of theory. In addition to absolute bond distances, Table 1 reports the spin-dependent bond-length difference ($\Delta l_{T\text{-}S}$), defined as the difference between singlet and triplet B-X bond lengths ($\Delta l_{T-S} = l_S^{B-X} - l_T^{B-X}$), thereby providing a direct structural measure of spin-state effects across different substituents.

As indicated in Table 1, the B–X bond lengths span a wide range depending on the nature of the substituent, reflecting differences in atomic size, bonding characteristics, and electronic interaction with the monovalent boron center. In the singlet state, short B–X distances are observed for light substituents such as H, F, OH, and NH_2, whereas heavier or more diffuse substituents, including SiH_3, PH_2, SH, and halogens beyond fluorine, lead to significantly elongated bonds.

A clear spin-dependent structural response is evident from the $\Delta l_{T\text{-}S}$ values. For several substituents most notably ethynyl, vinyl, phenyl, SiH_3, PH_2, and SH, the B–X bond is substantially shorter in the triplet state than in the singlet state, resulting in positive $\Delta l_{T\text{-}S}$ values exceeding 0.10 Å in some cases. This pronounced contraction upon spin excitation suggests enhanced bonding interactions in the triplet state for substituents capable of effective orbital overlap or spin delocalization.

In contrast, alkyl substituents such as Me, Et, iPr, and tBu exhibit very small or slightly negative $\Delta l_{T\text{-}S}$ values, indicating that the B–C bond length is largely insensitive to spin multiplicity in these systems. A similar trend is observed for strongly electronegative substituents such as F and OH, where the triplet state leads to modest bond elongation relative to the singlet. These observations imply that σ-donor-dominated substituents primarily stabilize the singlet bonding framework without inducing substantial spin-driven structural reorganization.

Overall, the data demonstrate that although geometric descriptors in monovalent borylenes are inherently limited to bond-length analysis, B–X distances and their spin-dependent variations capture clear and systematic substituent effects. The magnitude and sign of $\Delta l_{T\text{-}S}$ serve as an effective structural fingerprint of how different substituents encode spin-state preferences at the boron center, laying a solid foundation for the subsequent electronic structure and reactivity analyses presented in the following sections.

Table 1: Optimized B–X bond lengths (Å) for singlet and triplet borylenes and the corresponding spin-state-dependent bond-length differences ($\Delta l_{T\text{-}S}$)

R	B-X bond length (Å)		Δl_{T-S} (Å)
	Singlet state	Triplet state	
H	1.245	1.194	+0.051
Me	1.552	1.554	-0.002
Et	1.554	1.564	-0.010
ipr	1.557	1.568	-0.011
tBu	1.563	1.584	-0.021
Ethynyl	1.537	1.427	+0.110
Vinyl	1.548	1.455	+0.093
Ph	1.546	1.485	+0.061
SiH_3	2.154	1.991	+0.163
NH_2	1.387	1.374	+0.013
PH_2	1.992	1.828	+0.164
OH	1.307	1.346	-0.039
SH	1.841	1.729	+0.112
F	1.274	1.323	-0.049
Cl	1.744	1.721	+0.023
Br	1.908	1.865	+0.043

Singlet–Triplet Energetics of Substituted Borylenes

Table 2 reports the electronic energies including zero-point energy corrections for all sixteen borylene derivatives in both their singlet and triplet electronic states, as well as the corresponding $\Delta E_{S\text{-}T}$. The calculated energy differences provide a direct quantitative measure of spin-state stability, enabling systematic assessment of substituent-dependent preferences for singlet or triplet configurations [18].

The $\Delta E_{S\text{-}T}$ was calculated according to the following expression (Equation 1) [19]:

$$\Delta E_{S-T} = E_T^{elec+ZPE} - E_S^{elec+ZPE} \quad (1)$$

A positive value of ΔE_{S-T} indicates that the singlet state is energetically more stable than the triplet state, whereas a negative value would imply triplet-state stabilization.

Analysis of $E_S^{elec+ZPE}$ and $E_T^{elec+ZPE}$

The absolute electronic energies of both singlet and triplet states vary substantially across the series, primarily reflecting differences in substituent size, composition, and total electron count. As expected, derivatives containing heavier atoms such as Si, P, Cl, and Br exhibit markedly lower total energies compared to lighter substituents. While these absolute values are not directly comparable between different R groups, the consistent computational protocol ensures that relative energy differences within each derivative are meaningful and suitable for evaluating spin-state preferences.

Analysis of $\Delta E_{S\text{-}T}$

The $\Delta E_{S\text{-}T}$ values are positive for all investigated borylenes, indicating that the singlet state is energetically favored over the triplet state throughout the entire substituent set. However, the magnitude of the singlet–triplet gap shows pronounced substituent dependence, spanning a wide energy range from approximately 1.1×10^3 to nearly 8.0×10^3 kcal mol^{-1}.

Small $\Delta E_{S\text{-}T}$ values are observed for substituents such as SiH_3 and PH_2, suggesting relatively accessible triplet states and enhanced spin-state flexibility. In contrast, strongly electronegative substituents such as F, OH, and Cl lead to exceptionally large singlet–triplet gaps, reflecting strong singlet stabilization and a pronounced energetic penalty for triplet excitation. Alkyl and aryl substituents display intermediate $\Delta E_{S\text{-}T}$ values, indicating a balance between σ-donation and limited spin delocalization.

Table 2: Electronic energies corrected for zero-point vibrational energy ($E^{elec+ZPE}$) and ΔE_{S-T} of substituted borylenes (kcal mol^{-1})

R	$E_S^{elec+ZPE}$ (kcal mol^{-1})	$E_T^{elec+ZPE}$ (kcal mol^{-1})	ΔE_{S-T} (kcal mol^{-1})
H	-1586780.589	-1584188.164	+2592.425
Me	-4054680.626	-4051127.786	+3552.840
Et	-6520082.253	-6516773.297	+3308.956
ipr	-8985673.100	-8982288.080	+3385.020
tBu	-11451384.259	-11447937.419	+3446.840
Ethynyl	-6366984.250	-6364851.918	+2132.332
Vinyl	-6443863.180	-6441566.292	+2296.888
Ph	-16084694.365	-16081826.988	+2867.377
SiH_3	-19829331.613	-19828262.748	+1068.865
NH_2	-5065499.348	-5061135.774	+4363.574
PH_2	-23048220.538	-23046427.110	+1793.428
OH	-6314762.574	-6308139.015	+6623.559
SH	-26580976.366	-26578080.873	+2895.493
F	-7823331.839	-7815353.857	+7977.982
Cl	-30437100.515	-30431750.543	+5349.972
Br	-162954947.256	-162949964.681	+4982.575

Taken together, the results demonstrate that although all monovalent borylenes considered here are intrinsically singlet-ground-state species, the extent of singlet stabilization is highly sensitive to substituent identity. The wide variation in ΔE_{S-T} underscores the critical role of substituent-encoded electronic effects in modulating spin-state energetics. These findings establish ΔE_{S-T} as a robust energetic descriptor for assessing spin accessibility in borylenes and provide a quantitative foundation for correlating spin-state energetics with structural, electronic, and reactivity trends discussed in the subsequent sections.

Figure 1 illustrates the correlation between the ΔE_{S-T} and the B–X bond length difference (Δl_{T-S}) for the investigated monovalent borylene derivatives. The plot evaluates how structural variations upon spin-state switching relate to the relative stability of singlet and triplet states.

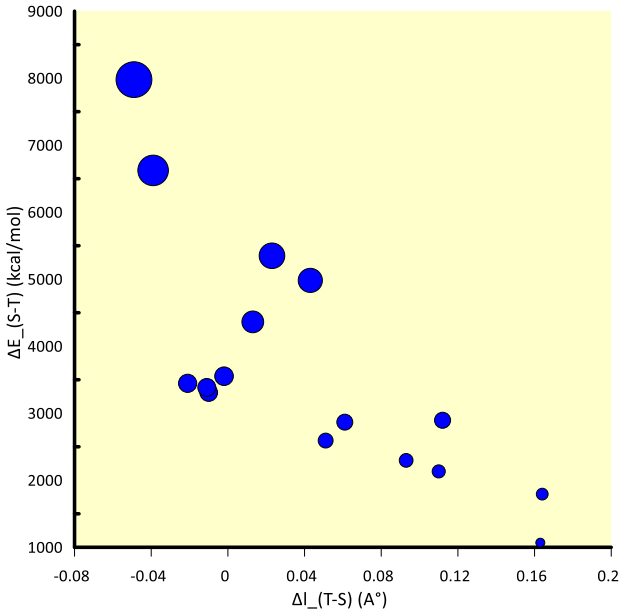


Figure 1: Correlation between the ΔE_{S-T} and the B–X bond length difference (Δl_{T-S}) in monovalent borylenes

Interpretation of the Signs

Energy Difference (ΔE_{S-T})

$\Delta E_{S-T} > 0$: The singlet state is more stable than the triplet state.
$\Delta E_{S-T} < 0$: The triplet state is more stable than the singlet state.

Bond Length Difference (Δl_{T-S})

$\Delta l_{T-S} > 0$: The B–X bond is longer in the singlet state than in the triplet state ($l_S > l_T$).
$\Delta l_{T-S} < 0$: The B–X bond is shorter in the singlet state than in the triplet state ($l_S < l_T$).
The plot reveals a clear inverse relationship between ΔE_{S-T} and Δl_{T-S}. As the bond length difference increases (moving toward positive Δl values), the ΔE_{S-T} decreases.
In systems where Δl_{T-S} is negative, the singlet state exhibits a shorter B–X bond, indicating stronger bonding interactions (likely enhanced π-donation or improved orbital overlap). These systems show larger positive ΔE_{S-T} values, meaning the singlet state is significantly more stabilized relative to the triplet.
Conversely, when Δl_{T-S} becomes positive, the singlet B–X bond is elongated compared to the triplet. This suggests weakened bonding interactions in the singlet state, which corresponds to a reduced ΔE_{S-T} [20].
Overall, Figure 1 demonstrates that structural variations in the B–X bond are directly linked to spin-state energetics. The degree of bond contraction or elongation upon spin transition plays a key role in governing the relative stability of singlet and triplet states in monovalent borylenes.

Electric Properties and Polarizability Features of Singlet and Triplet Borylenes

Polarizability is a fundamental electronic property that quantifies the ability of a molecular electron cloud to distort in response to an external electric field. It reflects the ease with which charge distribution can be redistributed under perturbation and therefore serves as a direct measure of electronic softness, delocalization capacity, and responsiveness. In low-coordinate main-group species such as monovalent borylenes, polarizability provides valuable insight into how substituents modulate electronic flexibility and spin-state sensitivity [21-25].
Table 3 compiles the calculated dipole moments (Debye), isotropic polarizabilities (α_{iso}), and anisotropic polarizabilities (α_{aniso}) for all sixteen derivatives in both singlet and triplet states. While dipole moments describe the permanent charge separation within each molecule, α_{iso} characterizes the overall magnitude of electronic deformability, and α_{aniso} reflects directional dependence of that deformation. Together, these parameters provide a comprehensive picture of molecular polarity and electronic adaptability across different substituents and spin states.
The isotropic polarizability (α_{iso}) is obtained as the average of the diagonal elements of the polarizability tensor (Equation 2) [26]:

$$\alpha_{iso} = \frac{\alpha_{xx} + \alpha_{yy} + \alpha_{zz}}{3} \quad (2)$$

The anisotropic polarizability (α_{aniso}) is calculated according to Equation 3 [8]:

$$\alpha_{aniso} = \sqrt{\frac{(\alpha_{xx} - \alpha_{yy})^2 + (\alpha_{yy} - \alpha_{zz})^2 + (\alpha_{zz} - \alpha_{xx})^2}{2}} \quad (3)$$

These definitions allow quantitative assessment of both the magnitude and directional character of electronic distortion.

Dipole Moments

The dipole moment values show a clear and systematic spin-state dependence. For nearly all derivatives, the singlet state exhibits significantly larger dipole moments than the corresponding triplet state. This trend indicates stronger intrinsic charge separation in the singlet configuration, consistent with a more polarized bonding framework.
π-Conjugated substituents such as ethynyl, vinyl, and phenyl display relatively large singlet dipole

moments, suggesting enhanced charge redistribution along the B–R axis. In contrast, heavier substituents such as SiH_3 and Br show very small triplet dipole moments, indicating a more symmetric or less polarized electronic distribution in that spin state.

Notably, heteroatom-containing substituents (NH_2, OH) retain appreciable dipole moments even in the triplet state, reflecting persistent polarity arising from electronegativity differences.

Table 3: Calculated dipole moments (Debye), isotropic polarizabilities (α_{iso}, Å^3), and polarizability anisotropies (α_{aniso}, Å^3) of singlet and triplet borylenes

R	Dipole moment (Debye)		α_{iso} (Å^3)		α_{aniso} (Å^3)	
	Singlet state	Triplet state	Singlet state	Triplet state	Singlet state	Triplet state
H	1.289	0.172	2.425	1.553	0.332	0.645
Me	2.515	0.877	3.878	3.291	0.271	1.187
Et	2.589	0.859	5.413	5.012	0.730	1.383
ipr	2.587	0.860	6.944	6.714	0.794	1.568
tBu	2.547	0.924	8.451	8.332	0.183	0.976
Ethynyl	3.655	0.860	4.716	4.164	4.289	5.849
Vinyl	3.045	0.382	5.184	4.907	3.245	4.874
Ph	3.753	0.122	10.783	10.670	8.812	10.741
SiH_3	0.999	0.092	6.115	8.436	2.971	8.224
NH_2	3.646	1.891	3.252	2.521	1.152	1.736
PH_2	1.269	0.948	5.761	5.083	3.076	3.502
OH	2.998	1.614	2.712	1.973	0.538	0.996
SH	1.614	0.883	4.586	3.951	2.900	3.022
F	1.042	0.693	2.206	1.437	0.209	0.433
Cl	1.022	0.366	3.495	2.689	2.356	2.378
Br	1.151	0.052	4.453	3.690	3.181	2.894

Isotropic Polarizability (α_{iso})

The isotropic polarizability values increase markedly with substituent size and electron count. As expected, bulky and heavier groups such as tBu, phenyl, SiH_3, PH_2, and Br exhibit larger α_{iso} values, reflecting greater overall electronic deformability.

In most cases, α_{iso} is slightly larger in the singlet state than in the triplet state; however, notable exceptions exist (*e.g.,* SiH_3), where the triplet state shows enhanced isotropic polarizability. This suggests that spin excitation can, in some systems, increase electronic flexibility rather than simply reduce it.

The particularly large α_{iso} values for phenyl derivatives highlight the contribution of extended π-systems to overall molecular softness.

Anisotropic Polarizability (α_{aniso})

The anisotropic component shows stronger sensitivity to substituent identity and spin state. Conjugated substituents such as phenyl, ethynyl, and vinyl exhibit markedly high α_{aniso} values, indicating significant directionality in their electronic response. This is consistent with delocalized π-electron systems, which inherently possess anisotropic charge distribution.

Interestingly, for many derivatives the triplet state displays higher α_{aniso} values than the singlet state. This suggests that unpaired electrons in the triplet configuration introduce directional asymmetry into the electronic cloud, enhancing anisotropic responsiveness.

Alkyl substituents, by contrast, show comparatively small anisotropy, reflecting their more isotropic σ-framework.

Table 3 demonstrates that polarity and polarizability in monovalent borylenes are strongly modulated by both substituent identity and spin state. The singlet state generally exhibits greater permanent polarity, whereas the triplet state often enhances directional electronic flexibility, particularly in conjugated systems. Larger substituents systematically

increase isotropic polarizability, while π-systems amplify anisotropic response.

Collectively, these findings indicate that substituents do not merely alter static structure but actively encode the electronic responsiveness of the boron center. The combined analysis of dipole moments and polarizabilities, thus, provides essential insight into how spin state and substituent effects cooperate to shape the reactivity landscape of monovalent borylenes.

Figure 2 presents the relationship between the dipole moment (μ) and the isotropic polarizability (α_{iso}) for the investigated borylene derivatives in both singlet and triplet states. The left panel corresponds to the singlet state, while the right panel represents the triplet state.

Figure 2 evaluates how molecular polarity is related to the overall electronic deformability of the system and how this relationship changes upon spin-state transition.

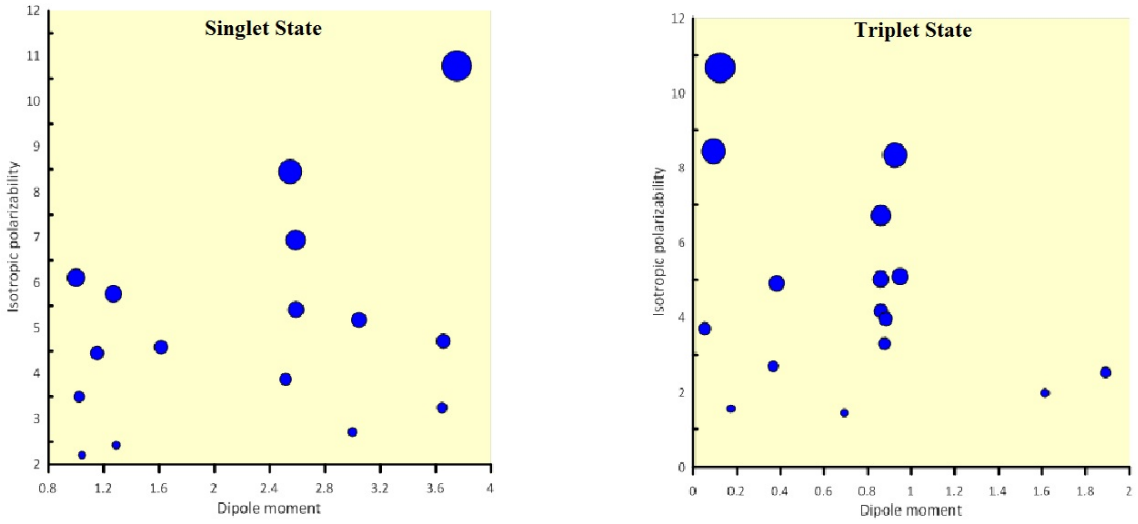


Figure 2: Relationship between dipole moment (μ) and isotropic polarizability (α_{iso}) in the singlet and triplet states of monovalent borylene derivatives

Singlet State

In the singlet state, a broad distribution of dipole moments is observed, ranging approximately from low to relatively high μ values. The isotropic polarizability shows noticeable dispersion and tends to increase in systems with larger substituents or more extended electronic structures.

Although a strict linear correlation is not evident, there is a general tendency for systems with higher dipole moments to exhibit moderate to higher isotropic polarizabilities. This behavior suggests that, in the singlet state, increased charge separation is often accompanied by enhanced electronic deformability. The presence of donor–acceptor interactions and π-conjugation likely contributes to both increased polarity and polarizability.

Triplet State

In the triplet state, the dipole moments are generally lower compared to the singlet state, and the data points appear more clustered in the low-to-intermediate μ region. The isotropic polarizability values also display a somewhat narrower distribution.

The reduced dipole moments in the triplet state can be attributed to the open-shell electronic configuration, which may lead to more symmetric or less polarized charge distributions in certain derivatives. The relationship between μ and α_{iso} appears weaker and less structured than in the singlet state, indicating that polarity plays a less dominant role in governing electronic deformability in the triplet configuration.

Comparison of the two panels reveals that:

- The singlet state generally exhibits higher dipole moments than the triplet state.

- The singlet state shows a broader variation in isotropic polarizability.
- The apparent coupling between polarity and polarizability is more pronounced in the singlet state than in the triplet state.

These observations suggest that electronic redistribution upon spin transition significantly affects molecular polarity, which in turn influences overall electronic softness. In singlet borylenes, polar effects and bonding interactions appear to enhance both dipole moment and electronic flexibility. In contrast, the triplet state reflects a more radical-like electronic structure with comparatively reduced polarity dependence.

Overall, Figure 2 demonstrates that the interplay between polarity and electronic deformability is spin-state dependent in monovalent borylenes.

Charge Distribution and B–X Bond Polarity in Singlet and Triplet Borylenes

Table 4 presents the Mulliken atomic charges for the boron atom (q_B) and the substituent atom directly bonded to boron (q_X) in both singlet and triplet states. Additionally, Table 4 reports the corresponding bond polarity parameter (Δq_{B-X}), defined as the absolute charge difference between the bonded atoms.

These data provide insights into how electron density is distributed across the B–X bond and how substituent identity and spin multiplicity influence charge separation. Although Mulliken charges are basis-set dependent, they remain valuable for identifying systematic trends within a consistently treated series, as is the case here.

Boron Atomic Charge (q_B)

In the singlet state, boron generally carries a small positive charge for alkyl and halogen substituents (*e.g.,* Me, Et, F, Cl, and Br), reflecting electron withdrawal from the electron-deficient boron center. However, for π-conjugated substituents such as ethynyl and phenyl, boron becomes slightly negative, indicating partial back-donation or charge delocalization toward the boron atom.

Table 4: Mulliken charge distribution and B–X bond polarity (Δq_{B-X}) in singlet and triplet borylenes

R	q_B		q_X		Δq_{B-X}	
	Singlet state	Triplet state	Singlet state	Triplet state	Singlet state	Triplet state
H	0.075	-0.014	-0.075	0.014	0.150	0.028
Me	0.073	0.039	-0.499	-0.429	0.572	0.468
Et	0.060	0.023	-0.343	-0.286	0.403	0.309
ipr	0.052	0.005	-0.198	-0.136	0.250	0.141
tBu	0.051	0.006	-0.076	-0.014	0.127	0.020
Ethynyl	-0.046	-0.116	-0.052	0.076	0.006	0.192
Vinyl	0.008	0.069	-0.168	-0.120	0.176	0.189
Ph	-0.034	0.023	-0.011	0.088	0.023	0.065
SiH_3	0.000	-0.047	0.204	0.278	0.204	0.325
NH_2	0.007	-0.007	-0.552	-0.514	0.559	0.507
PH_2	-0.002	-0.018	-0.026	0.000	0.024	0.018
OH	0.013	0.057	-0.346	-0.382	0.359	0.439
SH	-0.023	-0.016	-0.075	-0.043	0.052	0.027
F	0.097	0.173	-0.097	-0.173	0.194	0.346
Cl	0.043	0.051	-0.043	-0.051	0.086	0.102
Br	0.111	0.082	-0.111	-0.082	0.222	0.164

In the triplet state, notable charge redistribution occurs. For several substituents (*e.g.,* F, vinyl, and OH), boron becomes more positive compared to the singlet state, suggesting reduced electron donation in the open-shell configuration. Conversely, for SiH_3 and PH_2, boron becomes more negative, implying enhanced electron density localization at boron upon spin excitation.

Overall, the data show that spin-state changes can significantly alter the electronic character of

the boron center, even when geometric changes are modest.

Substituent Atomic Charge (q_X)

The substituent charges reveal complementary behavior. Electronegative substituents such as NH_2, OH, F, and halogens generally carry negative charge in both spin states, consistent with their electron-withdrawing nature. Alkyl substituents also display negative charge, reflecting polarization of the B–C bond toward carbon.

Interestingly, in some triplet configurations (*e.g.,* ethynyl and phenyl), the substituent charge becomes less negative or even slightly positive relative to the singlet state, indicating altered charge delocalization patterns in the open-shell electronic structure.

For SiH_3, the substituent bears positive charge in both states, consistent with silicon's lower electronegativity relative to boron and the resulting bond polarization toward boron.

Bond Polarity (Δq_{B-X})

The Δq_{B-X} values provide a direct quantitative measure of bond polarity. Large values are observed for strongly electronegative substituents such as NH_2, Me, F, and OH, indicating substantial charge separation across the bond.

In general, Δq_{B-X} tends to increase in the triplet state for several substituents (*e.g.,* F, OH, and SiH_3), suggesting enhanced charge polarization upon spin excitation. However, for bulky alkyl groups (iPr and tBu), the polarity decreases in the triplet state, indicating a more evenly distributed electron density.

The smallest polarity values are observed for π-conjugated substituents such as ethynyl and phenyl in the singlet state, consistent with partial charge delocalization reducing localized bond polarization.

Overall Conclusion

The Mulliken population analysis reveals that both substituent identity and spin multiplicity exert a pronounced influence on charge distribution within monovalent borylenes. While the singlet state often exhibits more localized and substituent-dependent polarization patterns, the triplet state can either enhance or attenuate bond polarity depending on the electronic nature of the substituent.

These findings demonstrate that spin-state changes do not merely alter energy and geometry but significantly reshape electron density distribution at the boron center. Consequently, charge analysis supports the concept that substituents encode distinct electronic signatures into the B–X bond, reinforcing the broader framework of spin-dependent reactivity modulation in monovalent boron systems.

Figure 3 illustrates the relationship between the Mulliken charge difference on the B–X bond (Δq_{B-X}) and the anisotropic polarizability (α_{aniso}) for the investigated borylene derivatives in both singlet and triplet states. The left panel corresponds to the singlet state, while the right panel represents the triplet state.

Figure 3 examines how charge redistribution along the B–X bond influences the directional electronic response of the molecule, and how this relationship depends on the spin state.

Singlet State

In the singlet state, the data points show a relatively scattered distribution. Systems with small Δq_{B-X} values (*i.e.,* limited charge separation along the bond) can still exhibit large anisotropic polarizabilities, particularly in cases involving π-delocalized or conjugated substituents. This suggests that in the singlet configuration, anisotropy is not governed solely by local charge accumulation, but also by orbital interactions and π-donation effects.

At moderate to higher Δq_{B-X} values, α_{aniso} generally remains within a lower-to-intermediate range, indicating that increasing bond polarity does not necessarily enhance directional electronic flexibility in the singlet state.

Triplet State

In contrast, the triplet state shows a more structured distribution. Certain systems with low to moderate Δq_{B-X} values exhibit

significantly higher anisotropic polarizabilities. This indicates that, in the open-shell configuration, directional electronic response is more sensitive to changes in charge distribution along the B–X bond.

Compared to the singlet state, the triplet plot suggests a stronger coupling between bond polarity and electronic anisotropy. The presence of unpaired electrons likely enhances directional polarizability, particularly in systems where charge redistribution facilitates spin delocalization.

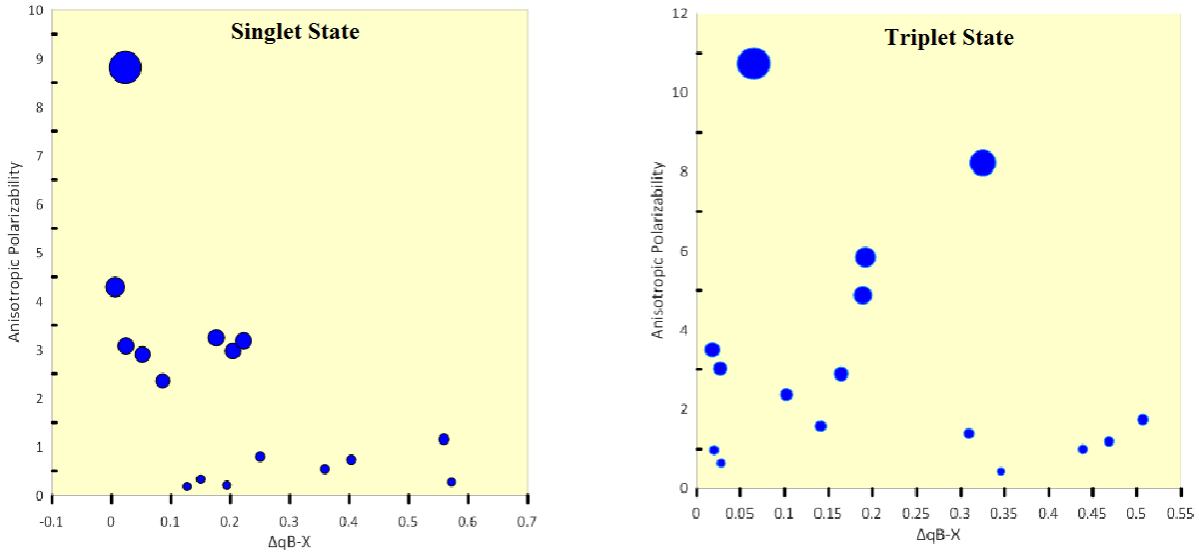


Figure 3: Correlation between Mulliken charge difference (Δq_{B-X}) and anisotropic polarizability (α_{aniso}) in the singlet and triplet states of monovalent borylenes

Comparison of the two panels reveals several important points:

- The singlet state displays weaker and less systematic correlation between Δq_{B-X} and α_{aniso}.
- The triplet state exhibits a more pronounced relationship between charge redistribution and anisotropic electronic response.
- Anisotropic polarizability appears more sensitive to spin-state effects than isotropic polarizability.

These observations suggest that, upon spin transition, the directional electronic flexibility of borylenes becomes more strongly influenced by bond polarity. In the singlet state, anisotropy is largely controlled by bonding interactions and π-structure, whereas in the triplet state, charge redistribution plays a more direct role in modulating electronic anisotropy.

Overall, Figure 3 demonstrates that the interplay between bond polarity and directional polarizability is spin-dependent in monovalent borylene derivatives.

Frontier Molecular Orbital Analysis and Spin-State Electronic Structure of Borylenes

Figure 4 shows the HOMO orbitals of the investigated borylene derivatives in both singlet and triplet states.

Orbital Analysis of Singlet State borylenes

In the singlet state, the HOMO is predominantly localized on the boron center and exhibits strong lone-pair character with noticeable σ-type symmetry. In several derivatives (especially π-donating substituents such as vinyl, ethynyl, and phenyl), partial delocalization toward the substituent is observed, indicating π-interaction between the boron center and the R group.

For electronegative substituents (F, Cl, and Br), the HOMO remains largely centered on boron, with limited back-donation, reflecting stronger σ-withdrawing effects.

Overall, the singlet HOMO reflects:

- LP localization at boron,

- Variable π-delocalization depending on substituent, and
- Closed-shell electronic structure.

Orbital Analysis of Triplet State Borylenes

In the triplet state, the HOMO exhibits clear open-shell character. The orbital density becomes more spatially separated and often displays p-type symmetry centered on boron. Compared to the singlet state:

- Electron density becomes more diffuse.
- Spin density distribution extends more significantly toward conjugated substituents.
- Orbital symmetry changes from lone-pair dominated to radical-like character.

This indicates that the triplet state promotes greater electronic redistribution and directional anisotropy.

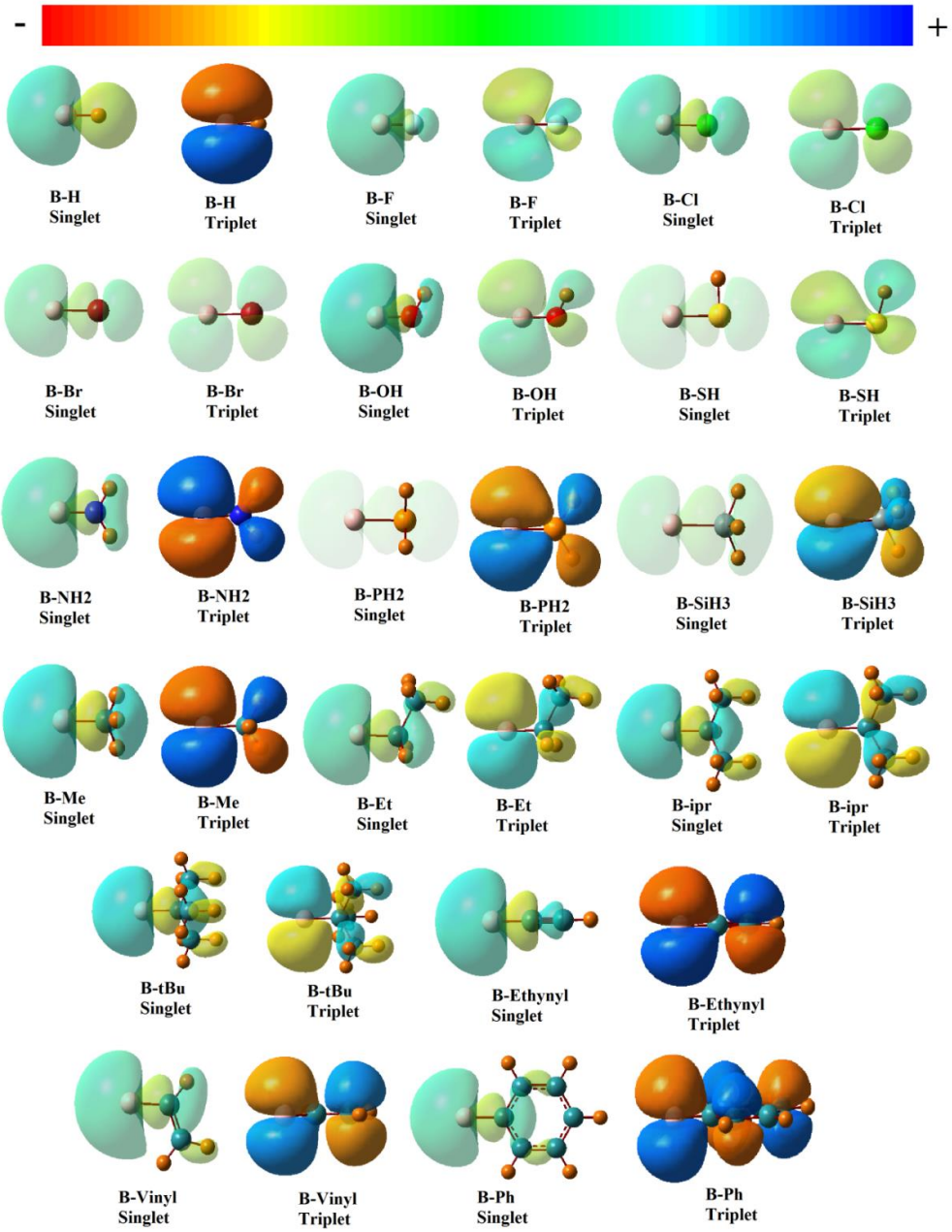


Figure 4: HOMO distributions of monovalent borylene derivatives in singlet and triplet states

The HOMO analysis demonstrates that:
- The singlet state is characterized by localized lone-pair stabilization.
- The triplet state exhibits radical-type p-orbital character at boron.
- π-donating substituents enhance orbital delocalization in both states but more strongly in the triplet state.

These observations support the idea that substituent effects modulate spin-state stability through orbital delocalization and electronic redistribution.

Table 5 presents the HOMO energies, LUMO energies, and HOMO–LUMO energy gaps (ΔE_S and ΔE_T) for both singlet and triplet states, along with the difference between singlet and triplet gaps ($\Delta E_S - \Delta E_T$). All energies are stated in eV units.

Table 5: HOMO and LUMO energies and HOMO–LUMO gaps of singlet and triplet borylene derivatives

R	Singlet state			Triplet state			$\Delta E_S - \Delta E_T$
	E_{HOMO}	E_{LUMO}	ΔE_S	E_{HOMO}	E_{LUMO}	ΔE_T	
H	-0.224	-0.075	0.149	-0.177	-0.085	0.092	0.057
Me	-0.200	-0.038	0.162	-0.145	-0.052	0.093	0.069
Et	-0.196	-0.040	0.156	-0.144	-0.053	0.091	0.065
ipr	-0.194	-0.038	0.156	-0.144	-0.051	0.093	0.063
tBu	-0.192	-0.036	0.156	-0.142	-0.050	0.092	0.064
Ethynyl	-0.216	-0.088	0.128	-0.174	-0.100	0.074	0.054
Vinyl	-0.199	-0.074	0.125	-0.165	-0.070	0.095	0.030
Ph	-0.194	-0.072	0.122	-0.153	-0.077	0.076	0.046
SiH_3	-0.205	-0.093	0.112	-0.189	-0.106	0.083	0.029
NH_2	-0.205	-0.028	0.177	-0.133	-0.010	0.123	0.054
PH_2	-0.211	-0.079	0.132	-0.186	-0.077	0.109	0.023
OH	-0.232	-0.016	0.216	-0.131	-0.020	0.111	0.105
SH	-0.233	-0.076	0.157	-0.181	-0.059	0.122	0.035
F	-0.269	-0.024	0.245	-0.144	-0.054	0.090	0.155
Cl	-0.260	-0.066	0.194	-0.167	-0.082	0.085	0.109
Br	-0.254	-0.070	0.184	-0.165	-0.083	0.082	0.102

In the singlet state:
- HOMO energies vary depending on substituent electron-donating or withdrawing ability.
- Strong electron-withdrawing substituents (F, Cl, and Br) stabilize the HOMO (more negative values).
- Donor groups reduce the HOMO–LUMO gap due to enhanced electron density and interaction.
- The singlet HOMO–LUMO gaps are generally larger than those of the triplet state, indicating higher electronic stability and lower reactivity for the singlet configuration.

In the triplet state:
- HOMO energies become less negative (destabilized relative to singlet).
- The HOMO–LUMO gaps are systematically smaller than in the singlet state.

This reflects:
- Open-shell character,
- Increased electronic softness, and
- Greater reactivity potential.

The reduced gap in the triplet state indicates enhanced polarizability and radical-like behavior.

The last column represents the difference between singlet and triplet HOMO–LUMO gaps.

Positive values indicate:
- The singlet gap is larger than the triplet gap.
- The triplet state is electronically softer.
- Larger values (*e.g.,* halogen substituents) suggest stronger substituent control over spin-dependent electronic structure.
- Smaller values (*e.g.,* π-conjugated groups) indicate closer electronic behavior between singlet and triplet states.

This parameter effectively measures spin-induced electronic reorganization.

Figure 5 presents the relationship between the ΔE_{S-T} and the difference in HOMO–LUMO energy

gaps between singlet and triplet states ($\Delta E_S - \Delta E_T$) (Equations 4 and 5).

$$\Delta E_S = E_S^{LUMO} - E_S^{HOMO} \quad (4)$$

$$\Delta E_T = E_T^{LUMO} - E_T^{HOMO} \quad (5)$$

The plot reveals a clear positive correlation:
- As the difference between singlet and triplet HOMO–LUMO gaps increases, the ΔE_{S-T} separation also increases.
- This indicates that spin-state stability is directly related to frontier orbital separation.

Systems with a large $\Delta E_S - \Delta E_T$ show:
- Stronger closed-shell stabilization in singlet,
- Greater energetic penalty for triplet formation, and
- Larger ΔE_{S-T} values.

Conversely, systems with smaller gap differences exhibit reduced singlet–triplet separation, reflecting increased electronic similarity between the two states.

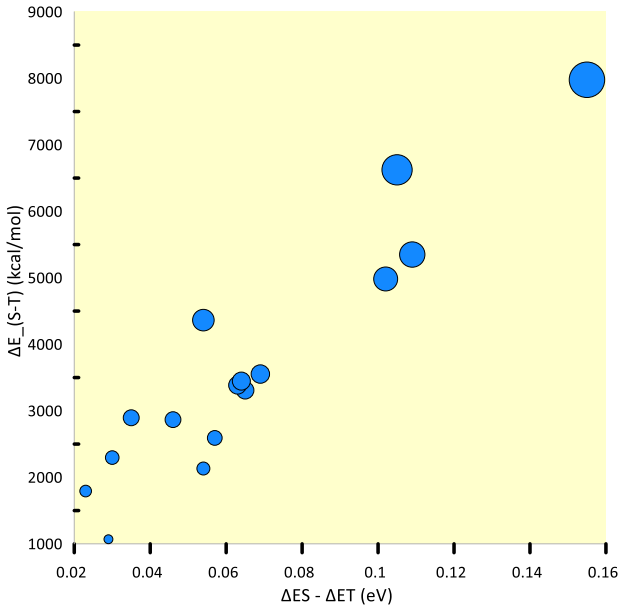


Figure 5: Correlation between ΔE_{S-T} and the difference in HOMO–LUMO gaps ($\Delta E_S - \Delta E_T$) in monovalent borylenes

The combined orbital and energetic analyses demonstrate that:
- Spin-state stability in borylenes is governed by frontier orbital interactions.
- Substituent effects modulate both HOMO localization and gap magnitude.
- The difference in HOMO–LUMO gaps serves as a predictive descriptor for ΔE_{S-T}.

This confirms that frontier molecular orbital engineering provides a rational strategy for tuning spin-state energetics in monovalent borylenes.

MEP Analysis of Singlet and Triplet Borylenes

The MEP represents the electrostatic interaction between the molecular charge distribution and a positive test charge placed in the surrounding space. It provides a three-dimensional visualization of electron density distribution and highlights regions susceptible to electrophilic or nucleophilic attack [26-29].

In the MEP surface [30]:
- Red regions indicate electron-rich (negative potential) areas.
- Blue regions indicate electron-deficient (positive potential) areas.
- Green/yellow regions correspond to intermediate potential values.

Figure 6 presents the MEP surfaces of the investigated monovalent borylene derivatives in both singlet and triplet states, mapped onto the electron density surface.

Across all derivatives, the boron center exhibits significantly less positive potential (pale blue region), consistent with its electron-deficient character. This confirms that boron acts as a potential electrophilic site in both spin states.

It should be noted that the small red region observed near the boron atom – particularly in the singlet state – arises from the presence of the localized LP on boron. However, the extent and distribution of electrostatic potential vary notably depending on both substituent type and spin state.

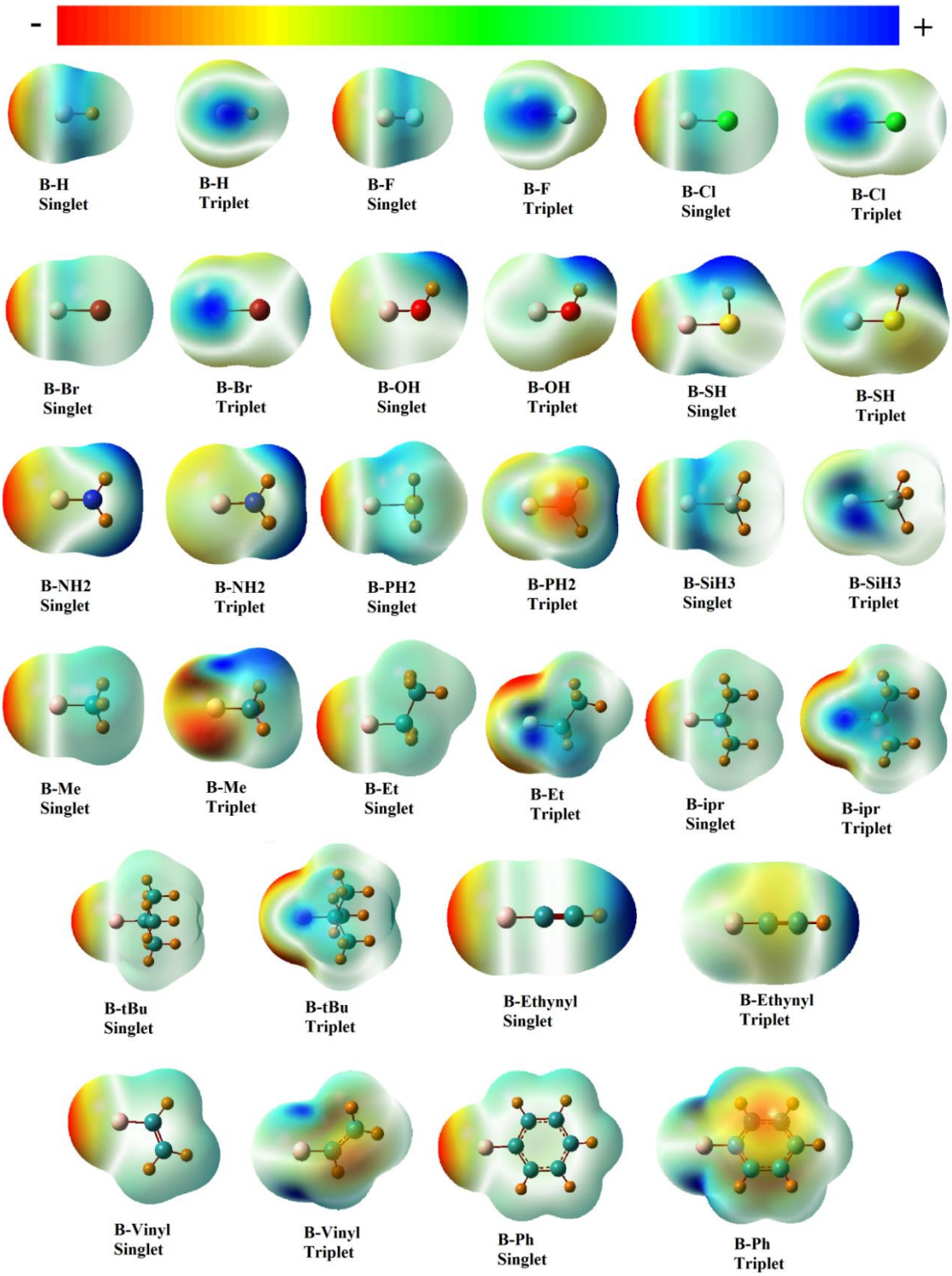


Figure 6: MEP surfaces of monovalent borylene derivatives in singlet and triplet states

In the singlet state:
- Boron does not behave as a purely electron-deficient center.
- The LP character on boron atom leads to partial electron enrichment.
- The electrostatic distribution is generally more polarized along the B–X bond axis.

This indicates stronger charge separation and directional bonding interactions in the singlet configuration.
In the triplet state:
- The electrostatic potential becomes more diffuse.
- The positive region around boron is often more extended.
- Charge redistribution appears less localized compared to the singlet state.

For π-conjugated substituents (vinyl, ethynyl, and phenyl), the triplet state shows broader delocalization of potential, reflecting increased radical character and spin density delocalization.
Comparison between singlet and triplet states reveals:
- The singlet state generally exhibits stronger localized electrostatic polarization.
- The triplet state shows greater spatial diffusion of potential.
- Spin-state transition leads to redistribution of electrostatic potential, particularly in conjugated systems.

This confirms that spin-state switching influences not only orbital structure and energy gaps, but also electrostatic reactivity patterns.
Overall, the MEP analysis demonstrates that:
- The boron center remains the primary electrophilic site in all derivatives.
- Substituents modulate electrostatic distribution through inductive and resonance effects.
- Spin-state transition alters the degree of charge localization and electrostatic anisotropy.

Therefore, electrostatic potential mapping supports the conclusion that spin-state energetics and substituent effects cooperatively govern the reactivity and electronic structure of monovalent borylenes.

Computed ^{11}B-NMR Shielding and Chemical Shift Analysis of Singlet and Triplet Borylenes

Table 6 presents the calculated ^{11}B-NMR shielding constants and corresponding chemical shifts (ppm) for the boron center in both singlet and triplet states of the studied molecules. The chemical shifts were referenced to B_2H_6, and a degeneracy of 1 was considered for all systems.
In the singlet states, the ^{11}B chemical shifts span a relatively wide range, reflecting the strong dependence of boron shielding on the electronic nature of the substituent. Electron-donating substituents generally lead to increased shielding (lower chemical shift values), whereas electron-withdrawing groups cause deshielding and higher ppm values. For instance, strongly electronegative substituents such as halogens show relatively downfield shifts compared to alkyl groups. The variation in singlet chemical shifts indicates that the boron center is highly sensitive to electronic redistribution and lone-pair donation effects in this spin state.
In the triplet states, the ^{11}B chemical shifts are significantly shifted downfield compared to the corresponding singlet values in most cases. This pronounced deshielding suggests a substantial change in the electronic environment of boron upon spin excitation. The presence of two unpaired electrons in the triplet state alters electron density distribution and reduces effective shielding at the boron nucleus. As a result, triplet chemical shifts are generally much larger (more positive ppm values) than those of the singlet counterparts, particularly for electronegative substituents.
Comparing singlet and triplet states overall, the triplet forms consistently exhibit stronger deshielding at boron relative to the singlet forms. This indicates that the singlet state maintains a comparatively higher localized electron density at the boron center. The difference between the two spin states becomes especially pronounced for highly electronegative substituents, highlighting the interplay between spin configuration and substituent electronic effects.
Figure 7 illustrates the relationship between the ΔE_{S-T} and the calculated ^{11}B chemical shifts for both spin states. In this plot, more positive and larger ΔE values correspond to greater singlet stability relative to the triplet state. The diagram reveals a general trend in which systems with larger positive ΔE_{S-T} values-indicating a more

stable singlet form-tend to cluster within specific chemical shift regions. In contrast, systems with smaller energy gaps exhibit more moderate shifts.

Table 6: Calculated ^{11}B-NMR shielding constants and chemical shifts of singlet and triplet borylenes (reference: B_2H_6)

R	Singlet state		Triplet state	
	^{11}B Shielding (ppm)	^{11}B Chemical shift (ppm)	^{11}B Shielding (ppm)	^{11}B Chemical Shift (ppm)
H	-202.595	297.498	-723.583	818.486
Me	-143.862	238.765	-500.294	595.197
Et	-144.736	239.639	-494.962	589.865
ipr	-141.130	236.033	-420.891	515.794
tBu	-134.452	229.355	-448.071	542.974
Ethynyl	-121.716	216.619	-798.871	893.774
Vinyl	-115.056	209.959	-135.177	230.080
Ph	-132.871	227.774	-411.425	506.328
SiH_3	-469.426	564.329	-764.686	859.589
NH_2	8.082	86.821	-14.940	109.843
PH_2	-239.812	334.715	-121.064	215.967
OH	66.657	28.246	-190.855	285.758
SH	-83.353	178.256	-72.434	167.337
F	85.677	9.226	-915.270	1010.173
Cl	-2.031	96.934	-949.768	1044.671
Br	-27.645	122.548	-968.076	1062.979

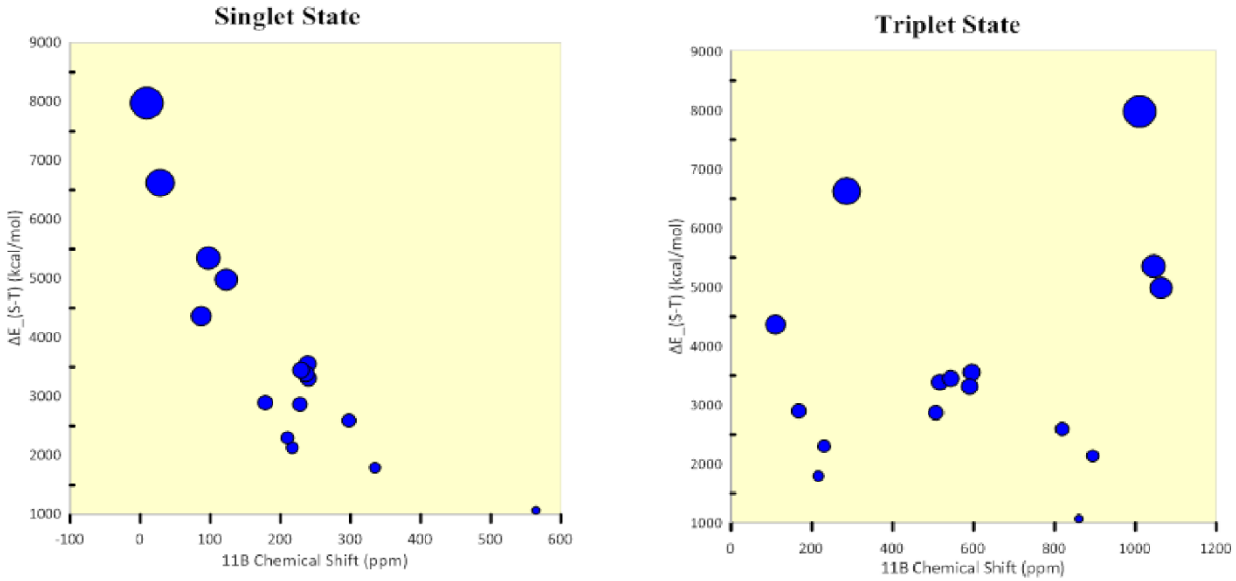


Figure 7: Correlation between ΔE_{S-T} and ^{11}B chemical shifts in the singlet and triplet states of monovalent borylenes

Overall, Figure 7 demonstrates a correlation between magnetic response (^{11}B chemical shift) and spin-state energetics. As the singlet state becomes increasingly stabilized (larger positive ΔE_{S-T}, characteristic changes in boron deshielding are observed. The data indicate that increased singlet stabilization correlates with enhanced shielding at the boron center, reflected by lower ^{11}B chemical shift values. This suggests that ^{11}B NMR parameters can serve as a sensitive probe of spin-state stability and electronic structure in these boron-containing systems.

Electronic Structure Analysis: Hybridization, Spin Density, and σ_{B-X} Bond Energies

Table 7 presents the natural hybridization of the boron atom in the B–X bond, the hybridization of the boron LP, the calculated spin density at boron in the triplet state, and the σ_{B-X} bond energies for both singlet and triplet states of the investigated borylene derivatives. These data provide detailed insight into the electronic structure variations as a function of substituent (R) and spin state.

Table 7: Natural hybridization of boron in the B–X bond, -LP hybridization, spin density (triplet), and σ_{B-X} bond energies for singlet and triplet states of borylene derivatives

R	B hybridization in B-X		B hybridization (LP)	B spin density	σ_{B-X} bond energy (kcal mol^{-1})	
	Singlet state	Triplet state	Singlet state	Triplet state	Singlet state	Triplet state
H	$sp^{6.86}d^{0.03}$	$sp^{3.53}d^{0.01}$	$sp^{0.14}$	1.910	-26024.668	-27721.698
Me	$sp^{4.54}d^{0.02}$	$sp^{2.59}d^{0.01}$	$pd^{0.01}$	1.892	-32923.870	-33075.749
Et	$sp^{4.57}d^{0.02}$	$sp^{2.61}d^{0.01}$	$sp^{0.20}$	1.875	-37768.938	-32513.420
ipr	$sp^{4.61}d^{0.02}$	$sp^{2.62}$	$sp^{0.20}$	1.863	-31417.003	-31973.684
tBu	$sp^{4.70}d^{0.02}$	$sp^{2.71}d^{0.01}$	$sp^{0.20}$	1.852	-30658.863	-31021.616
Ethynyl	$sp^{5.03}d^{0.03}$	$sp^{2.09}d^{0.01}$	$sp^{0.18}$	1.713	-37635.259	-40309.461
Vinyl	$sp^{4.69}d^{0.02}$	p	$sp^{0.20}$	1.573	-33162.358	-14879.756
Ph	$sp^{4.57}d^{0.02}$	$sp^{2.23}$	$sp^{0.20}$	1.637	-32765.715	-36106.427
SiH_3	$sp^{7.54}d^{0.03}$	$sp^{2.95}$	$sp^{0.13}$	1.772	-23896.479	-28459.127
NH_2	$pd^{0.01}$	$sp^{0.56}d^{0.01}$	$sp^{0.18}$	1.993	-19754.322	-18835.516
PH_2	$sp^{7.57}d^{0.04}$	$sp^{2.62}$	$sp^{0.13}$	1.811	-27181.334	-33797.488
OH	$pd^{0.01}$	$pd^{0.01}$	$sp^{0.15}$	1.952	-25365.061	-23528.078
SH	$pd^{0.01}$	$sp^{2.74}d^{0.01}$	$sp^{0.12}$	1.865	-19459.350	-41834.527
F	$pd^{0.01}$	$sp^{4.03}d^{0.02}$	$sp^{0.12}$	1.930	-32104.852	-64634.590
Cl	$pd^{0.01}$	$sp^{3.43}d^{0.01}$	$sp^{0.11}$	1.914	-24226.596	-46288.601
Br	$pd^{0.01}$	$pd^{0.01}$	$sp^{0.09}$	1.867	-21862.429	-21521.642

Boron Hybridization in the B–X Bond

The hybridization pattern of boron in the B–X bond exhibits clear differences between singlet and triplet states.

In the singlet state, boron generally shows higher p-character in many alkyl and aryl derivatives (*e.g.,* Me, Et, *i*Pr, *t*Bu, and Ph), often with significant sp^4-sp^7 contributions. In contrast, several electronegative substituents (NH_2, OH, F, Cl, and Br) display dominant p-character with minimal s-contribution, indicating a more directional bonding interaction.

In the triplet state, boron hybridization shifts toward lower p-exponents (*e.g.,* sp^2-sp^3 range for many derivatives), reflecting a more classical three-coordinate geometry with increased s-character in bonding. Notably, vinyl and halogen derivatives show substantial differences between singlet and triplet configurations, indicating that spin excitation significantly reorganizes the bonding orbital framework.

Overall, the triplet state generally exhibits a more balanced sp^2-sp^3-type hybridization, while the singlet state may display extreme p-dominance or unusually high p-exponents depending on substituent effects.

Hybridization of the Boron LP

The LP hybridization in the singlet state shows predominantly s-character with very small p-contribution (*e.g.,* $sp^{0.09}$-$sp^{0.2}$). The calculated LP hybridization indicates that the boron LP possesses predominantly s-character rather than p-character. The small exponent values confirm that the LP is strongly localized and energetically stabilized, which contributes significantly to singlet state stabilization.

Since the triplet state lacks a true LP (due to two unpaired electrons occupying separate orbitals),

the LP hybridization column mainly reflects singlet electronic structure. Therefore, this column emphasizes that singlet stabilization is closely associated with a localized s-type LP on boron.

Spin Density at Boron (Triplet State)

The spin density values in the triplet state range approximately from 1.57 to 1.99, indicating that the majority of unpaired electron density resides on the boron center.

Electron-donating substituents such as NH_2 and OH show spin densities close to 2.0, meaning the unpaired electrons are highly localized at boron.

In contrast, π-conjugated substituents (vinyl and phenyl) slightly reduce spin density, suggesting partial delocalization of spin density into the substituent framework.

Halogen substituents maintain relatively high spin density values (~1.9), indicating limited spin delocalization despite their strong electronegativity.

Overall, the triplet state remains predominantly boron-centered, with substituents modulating but not eliminating spin localization.

σ_{B-X} Bond Energies

The σ_{B-X} bond energies differ significantly between singlet and triplet states.

In several cases (*e.g.,* F, Cl, SH, and ethynyl), the triplet state shows substantially more negative σ-bond energies, indicating stronger σ-interactions in the triplet configuration. For example, F exhibits an exceptionally large stabilization in the triplet state.

Conversely, for some alkyl and amino substituents (*e.g.,* NH_2 and OH), the singlet state shows comparable or slightly stronger σ-bonding contributions.

These variations indicate that σ-bond strength is highly sensitive to both substituent electronegativity and spin state. Electronegative substituents particularly enhance σ-interactions in the triplet state, whereas electron-donating groups may favor singlet stabilization through lone-pair interactions rather than pure σ-strengthening.

Overall Conclusion

Table 7 demonstrates that spin-state switching significantly reorganizes the electronic structure at boron. The singlet state is characterized by a localized p-type LP and variable hybridization strongly influenced by substituent electronics, whereas the triplet state exhibits substantial spin localization at boron and often enhanced σ_{B-X} bonding for electronegative substituents. These findings highlight the delicate interplay between hybridization, spin distribution, and substituent effects in governing the stability and bonding characteristics of borylene derivatives.

Figure 8 illustrates the relationship between the ΔE_{S-T} and the σ_{B-X} bond energy for the investigated borylene derivatives in both spin states. The left panel corresponds to the singlet state, while the right panel represents the triplet state. In these plots, positive and larger ΔE_{S-T} values indicate greater thermodynamic stability of the singlet state relative to the triplet state.

Analysis of the Singlet State (Left Panel)

In the singlet state, a general trend can be observed between σ_{B-X} bond energy and ΔE_{S-T}. Derivatives with moderately strong σ_{B-X} interactions (more negative bond energies) tend to exhibit larger positive ΔE_{S-T} values, indicating enhanced singlet stabilization.

However, the relationship is not strictly linear. Some substituents with very strong σ_{B-X} interactions do not necessarily correspond to the highest ΔE_{S-T} values. This suggests that singlet stabilization is not governed solely by σ-bond strength, but also by additional electronic factors such as LP localization and π-donation from the substituent.

Overall, the singlet panel indicates that stronger σ_{B-X} bonding generally supports singlet stability, but it is not the only controlling factor.

Analysis of the Triplet State (Right Panel)

In the triplet state, the distribution of points is more scattered. Several derivatives exhibit very strong σ_{B-X} bond energies (particularly for electronegative substituents), yet their ΔE_{S-T} values vary significantly.

This indicates that strong σ-bonding in the triplet configuration does not necessarily translate into greater overall ΔE_{S-T} . In fact, in

some cases, enhanced σ-interactions in the triplet state may partially compensate for singlet stabilization.

Moreover, the broader spread of σ_{B-X} bond energies in the triplet panel suggests that substituent effects influence bonding more dramatically in the triplet configuration.

Comparing both panels reveals that:

In the singlet state, there is a clearer tendency for larger positive ΔE_{S-T} values to correlate with favorable σ_{B-X} bonding.

In the triplet state, the relationship between σ-bond strength and ΔE_{S-T} is less systematic.

Positive and larger ΔE_{S-T} values correspond to greater singlet stability, meaning derivatives located higher on the vertical axis are more singlet-favored.

The absence of a strict linear correlation in either panel indicates that singlet stabilization arises from a balance of σ-interactions, LP effects, and overall electronic structure reorganization rather than from σ-bond energy alone.

Thus, while σ_{B-X} bonding contributes to spin-state energetics, it does not exclusively determine singlet–triplet stability.

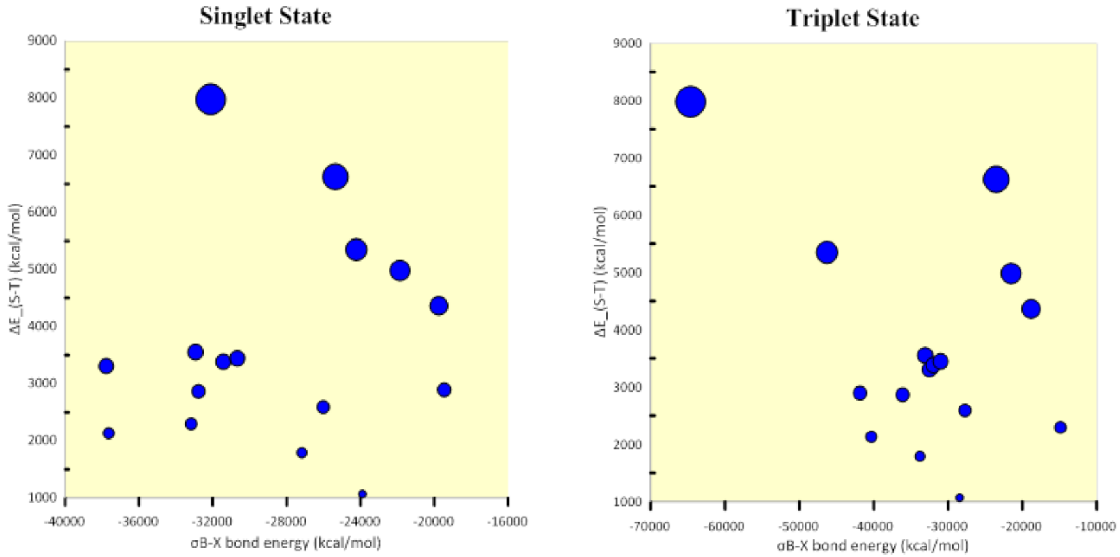


Figure 8: Correlation between ΔE_{S-T} and σ_{B-X} bond energy for borylene derivatives in singlet and triplet states

Conclusion

In this work, a comprehensive quantum chemical investigation was conducted on a systematically designed series of monovalent B–X derivatives in both singlet and triplet spin states to unravel the fundamental electronic factors governing their stability and reactivity. By integrating energetic, structural, magnetic, and response properties, a coherent image of substituent-controlled stabilization has been established. The calculated singlet–triplet energy separations clearly demonstrate that singlet stability is highly sensitive to the nature of the substituent. Larger positive ΔE_{S-T} values correspond to thermodynamically favored singlet states, and these systems consistently exhibit stronger σ_{B-X} bond energies. This direct correlation confirms that reinforcement of the σ-framework plays a decisive role in stabilizing the paired-electron configuration at boron. NBO analysis provides critical mechanistic insight into this stabilization. In stabilized singlet species, the boron LP displays increased s-character, indicating enhanced electron localization and reduced orbital diffuseness. This rehybridization effect emerges as a key electronic descriptor of singlet stabilization. In contrast, triplet states show pronounced spin density at the boron center, reflecting reduced pairing and altered bonding interactions. Magnetic properties further validate these electronic trends. The calculated ^{11}B-NMR chemical shifts systematically track changes in

electron density and bonding strength, confirming that greater singlet stabilization is associated with shielding behavior consistent with increased lone-pair localization. Variations in dipole moments reveal that substituent-induced electronic asymmetry significantly influences charge distribution, while isotropic and anisotropic polarizabilities demonstrate that electronically softer and more delocalized systems possess enhanced charge deformability. Importantly, the strong interdependence observed among $\Delta E_{S\text{-}T}$, $\sigma_{B\text{-}X}$ bond strength, lone-pair hybridization, spin density, NMR response, dipole moment, and polarizability establishes a unified structure–property relationship for B-centered systems. Rather than isolated trends, these properties evolve coherently under substituent control. Overall, this study provides a quantitative electronic framework for predicting singlet stabilization and tuning bonding characteristics in boron-centered species. The correlations identified here offer practical guidelines for the rational design of boron-containing reactive intermediates, ligands, and functional materials with tailored electronic and magnetic properties.

ORCID

Mehdi Nabati
https://orcid.org/0000-0002-3353-0628

References

[1] Chen, X., Yang, Y., Wang, H., Mo, Z. Cooperative bond activation and catalytic CO_2 functionalization with a geometrically constrained bis(silylene)-stabilized borylene. *Journal of the American Chemical Society*, **2023**, 145, 7011-7020.

[2] Fan, J., Pan, S., Yao, S., Ding, C., Frenking, G., Drieß, M. From bis(borylene)-substituted xanthenes as reactive intermediates to diboraoxirane complexes. *Journal of the American Chemical Society*, **2025**, 147, 6925-6933.

[3] Nazish, M., Legendre, C.M., Ding, Y., Schluschaß, B., Schwederski, B., Herbst-Irmer, R., Parvathy, P., Parameswaran, P., Stalke, D., Kaim, W., Roesky, H.W. A neutral borylene and its conversion to a radical by selective hydrogen transfer. *Inorganic Chemistry*, **2023**, 62, 9343-9349.

[4] Hu, J., Wang, X. Metalized borylene: A new inorganic cousin of borylenes. *ChemPhysChem*, **2025**, 26, e202401163.

[5] Soleilhavoup, M., Bertrand, G. Borylenes: An emerging class of compounds. *Angewandte Chemie International Edition*, **2017**, 56, 10282-10292.

[6] Zhang, C., Gilliard, R.J., Cummins, C.C. Arene extrusion as an approach to reductive elimination at boron: implication of carbene-ligated haloborylene as a transient reactive intermediate. *Chemical Science*, **2024**, 15, 17873-17880.

[7] Duret, G., Quinlan, R., Bisseret, P., Blanchard, N. Boron chemistry in a new light. *Chemical Science*, **2015**, 6, 5366-5382.

[8] Woods, W.G. An introduction to boron: history, sources, uses, and chemistry. *Environmental Health Perspectives*, **1994**, 102, 5-11.

[9] DeFrancesco, H., Dudley, J., Coca, A. Boron chemistry: An overview. *ACS symposium series*, **2016**, 1-25.

[10] Huang, Z., Wang, S., Dewhurst, R.D., Ignat'ev, N.V., Finze, M., Braunschweig, H. Boron: Its role in energy-related processes and applications. *Angewandte Chemie International Edition*, **2019**, 59, 8800-8816.

[11] Coghi, P., Zhu, Y., Xie, H., Hosmane, N.S., Zhang, Y. Organoboron compounds: Effective antibacterial and antiparasitic agents. *Molecules*, **2021**, 26, 3309.

[12] Messner, K., Vuong, B., Tranmer, G.K. The boron advantage: The evolution and diversification of boron's applications in medicinal chemistry. *Pharmaceuticals*, **2022**, 15, 264.

[13] Baker, S.J., Ding, C.Z., Akama, T., Zhang, Y.K., Hernandez, V., Xia, Y. Therapeutic potential of boron-containing compounds. *Future Medicinal Chemistry*, **2009**, 1, 1275-1288.

[14] Prieto-García, C.K., Quitián-Lara, H.M., Masqué, J.M., Fantuzzi, F., Jiménez-Halla, J.Ó.C. Can boronyl-bearing carbon chains serve as gas-phase molecular carriers of interstellar boron?. *ACS Earth and Space Chemistry*, **2025**, 9, 2045-2055.

[15] Núñez, R., Romero, I., Teixidor, F., Viñas, C. Icosahedral boron clusters: A perfect tool for the enhancement of polymer features. *Chemical Society Reviews*, **2016**, 45, 5147-5173.

[16] Curran, D.P., Boussonnière, A., Geib, S.J., Lacôte, E. The parent borylene: Betwixt and between. *Angewandte Chemie International Edition*, **2012**, 51, 1602-1605.

[17] Nabati, M. Decoding substituent-controlled singlet-triplet signatures of silylenes: A DFT study on divalent silicon species. *Journal of Pharmaceutical Sciences and Computational Chemistry*, **2026**, 2(1), 75-90.

[18] Ollivier, C., Renaud, P. Organoboranes as a Source of Radicals. *Chemical Reviews*, **2001**, 101, 3415-3434.

[19] Braunschweig, H., Rais, D., Uttinger, K. Terminal borylene complexes stabilized by a transition-metal

base. *Angewandte Chemie International Edition*, **2005**, 44, 3763-3766.
[20] Soleilhavoup, M., Bertrand, G. Borylene: Eine aufstrebende verbindungsklasse. *Angewandte Chemie*, **2017**, 129, 10416-10426.
[21] Sharmila, D., Mondal, B., Ramalakshmi, R., Kundu, S., Varghese, B., Ghosh, S. First-row transition-metal–diborane and –borylene complexes. *Chemistry - A European Journal*, **2015**, 21, 5074-5083.
[22] Hajipour, M., Ghiasi, R., Baniyaghoob, S. Insights into fluorine substituent effect on the electronic, optical and spectroscopic properties of borazanaphthalene. *Results in Chemistry*, **2025**, 15, 102326.
[23] Nabati, M., Bodaghi-Namileh, V. Evaluation of medicinal effects of isoxazole ring isosteres on zonisamide for autism treatment by binding to potassium voltage-gated channel subfamily d member 2 (Kv 4.2). *Advanced Journal of Chemistry-Section A*, **2020**, 3, 462-472.
[24] Nabati, M., Pournamdari, E., Bodaghi-Namileh, V., Dashti-Rahmatabadi, Y., Sarshar, S. Substitution of carbonyl group of ellagic acid with silanediol group for better inhibition of VEGFR-2 kinase enzyme. *Advanced Journal of Chemistry, Section B: Natural Products and Medical Chemistry*, **2020**, 2(2), 64-72.
[25] Shavitt, I. Geometry and singlet-triplet energy gap in methylene: A critical review of experimental and theoretical determinations. *Tetrahedron*, **1985**, 41, 1531-1542.
[26] Ghiasi, R., Amini, E. Substituent and solvent effects on geometric and electronic structure of $C_5H_5Ir(PH_3)_3$ iridabenzene: A theoretical insight. *Journal of Structural Chemistry*, **2015**, 56, 1483-1494.
[27] Ghiasi, R., Pasdar, H., Fereidoni, S. Solvent and substitution effects on the structure and properties of a half-sandwich complex of vanadium with a terminal borylene ligand: Theoretical study. *Russian Journal of Inorganic Chemistry*, **2016**, 61, 327-333.
[28] Soleimani-Amiri, S., Khanmohammadi, A., Vessally, E., Makasana, J., Ballal, S., Panigrahi, R., Joshi, K.K. Metal–porphyrin functionalized carbon nanocones as promising $COCl_2$ sensors: a DFT and NBO perspective. *Computational and Theoretical Chemistry*, **2025**, 1253, 115457.
[29] Haaf, M., Schmedake, T.A., West, R. Stable silylenes. *Accounts of Chemical Research*, **2000**, 33, 704-714.
[30] Hung, V., Korobkov, I., Nikonov, G.I. Half-sandwich silane σ-complexes of ruthenium supported by NHC carbene. *Organometallics*, **2016**, 35, 936-942.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.